

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 10 • Radians

With Answers

Questions with full worked solutions

- 57 pages
- Past papers up to June 2026

Radians

Your classified route

19
questions

173
marks

4
families

12
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Radians

Families and variants

F01 • Sector measures and inverse sector problems

V01 Direct arc, area and perimeter of one sector

V02 Recover radius or angle from arc length or sector area

V03 Concentric, annular and linked sectors

V04 Coupled Sector Area-and-Perimeter Constraints

F02 • Recovering the central angle and choosing the correct arc

V01 Angle from straight lines, symmetry or rotational partition

V02 Angle from triangle trigonometry

V03 Major, minor and reflex sector selection

F03 • Composite sector designs

V01 Minor sector joined to one or more polygons

V02 Major sector with an attached or removed polygon

V03 Repeated rotational sector design

F04 • Circular segments and chords

V01 Segment area

V02 Segment perimeter and chord

Sector measures and inverse sector problems

WHAT THIS FAMILY ASKS

Apply $s = r \theta$ and $A = \frac{1}{2} r^2 \theta$ directly or inversely, including linked and concentric sectors.

VARIANT ROUTE

V02 Recover radius or angle from arc length or sector area

V03 Concentric, annular and linked sectors

V04 Coupled Sector Area-and-Perimeter Constraints

5 Complete Classified questions

37 total marks

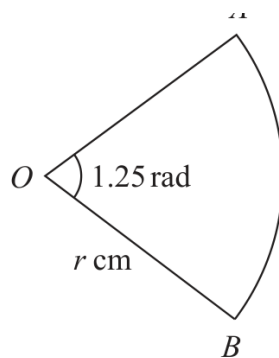


Figure 1

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is 1.25 radians.

Given that the area of the sector AOB is 15 cm^2

- (a) find the exact value of r , (2)
- (b) find the exact length of the perimeter of the sector. Write your answer in simplest form. (3)

(Total 5 marks)

Question 1

WMA11/01 October 2019

T10 Radians

F01 Sector measures and inverse sector problems

V02 Recover radius or angle from arc length or sector area

1

Use the sector area formulaFor a sector, $A = \frac{1}{2}r^2\theta$. Here $A = 15$ and $\theta = 1.25 = \frac{5}{4}$, so $15 = \frac{1}{2}r^2 \cdot \frac{5}{4}$.

2

Find the exact radius $r^2 = \frac{30}{5/4} = 24$, so $r = \sqrt{24} = 2\sqrt{6}$ because the radius is positive.

3

Find the perimeterArc length is $s = r\theta = (2\sqrt{6})\left(\frac{5}{4}\right) = \frac{5\sqrt{6}}{2}$. The perimeter is $2r + s = 4\sqrt{6} + \frac{5\sqrt{6}}{2} = \frac{13\sqrt{6}}{2}$.

FINAL ANSWER

(a) $r = 2\sqrt{6}$ cm. (b) $\frac{13\sqrt{6}}{2}$ cm.

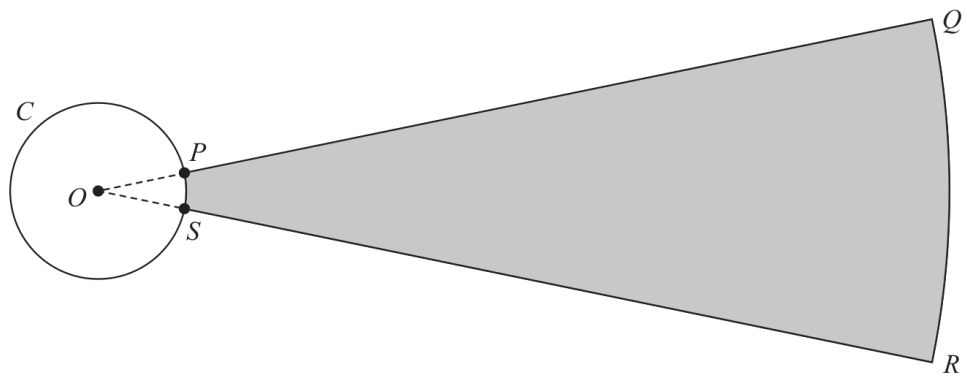


Figure 3

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2$ m

(1)

Given also that

- the target area, $PQRS$, is 90 m^2
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0$$

(3)

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

(Total for Question 9 is 7 marks)

Question 2

WMA11/01 October 2023

T10 Radians

F01 Sector measures and inverse sector problems

V03 Concentric, annular and linked sectors

1

Use arc length $s = r\theta$, so $0.72 = OP \times 0.6$. Hence $OP = 1.2$ m.

2

Set up the target areaThe outer sector has radius $x + 1.2$ and the inner sector has radius 1.2, with angle 0.6 radians.

3

Use sector area

$$\frac{1}{2}(x + 1.2)^2(0.6) - \frac{1}{2}(1.2)^2(0.6) = 90.$$

4

Simplify to the given quadratic
 $0.3((x + 1.2)^2 - 1.44) = 90$, so $0.3x^2 + 0.72x - 90 = 0$. Multiplying by $\frac{50}{3}$ gives $5x^2 + 12x - 1500 = 0$.

5

Find the positive value of x

$$x = \frac{-12 + \sqrt{12^2 + 4(5)(1500)}}{10} = \frac{-6 + 4\sqrt{471}}{5} \approx 16.16.$$

6

Find the perimeterPerimeter = $PQ + SR + \text{arc } QR + \text{arc } PS = x + x + 0.6(x + 1.2) + 0.72 \approx 43.46$ m.

7

Round to the nearest metre

The total perimeter is 43 m to the nearest metre.

FINAL ANSWER

(a) $OP = 1.2$ m. (b) $5x^2 + 12x - 1500 = 0$. (c) 43 m.

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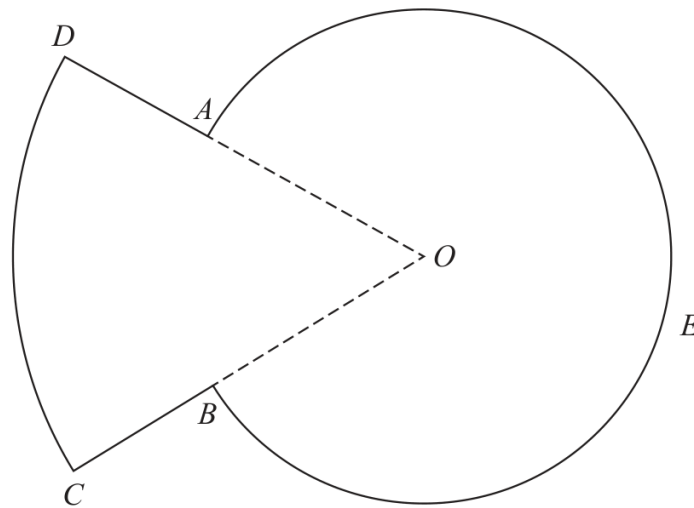


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA : AO = 3 : 5$

(a) show that $AO = 7.03$ m to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

(Total for Question 8 is 9 marks)

Question 3

WMA11/01 January 2024

T10 Radians

F01 Sector measures and inverse sector problems

V03 Concentric, annular and linked sectors

1

Find the outer radius

For sector DOC , arc length $CD = 9$ and angle 0.8 radians, so $OD = \frac{9}{0.8} = 11.25$ m.

2

Use the ratio DA to AO

$DA : AO = 3 : 5$, so $OD = DA + AO$ represents 8 parts. Hence $AO = \frac{5}{8}(11.25) = 7.03125$ m ≈ 7.03 m.

3

Find the straight side lengths

$DA = CB = OD - AO = 11.25 - 7.03125 = 4.21875$ m.

4

Find the major arc

The major sector $AOBEA$ has angle $2\pi - 0.8$, so arc $AEB = 7.03125(2\pi - 0.8)$.

5

Calculate the perimeter

Perimeter $= 9 + 2(4.21875) + 7.03125(2\pi - 0.8) \approx 56.0$ m.

6

Calculate the total area

Area $= \frac{1}{2}(11.25)^2(0.8) + \frac{1}{2}(7.03125)^2(2\pi - 0.8) \approx 186$ m² to the nearest whole number.

FINAL ANSWER

(a) $AO = 7.03$ m. (b) 56.0 m. (c) 186 m².

1.

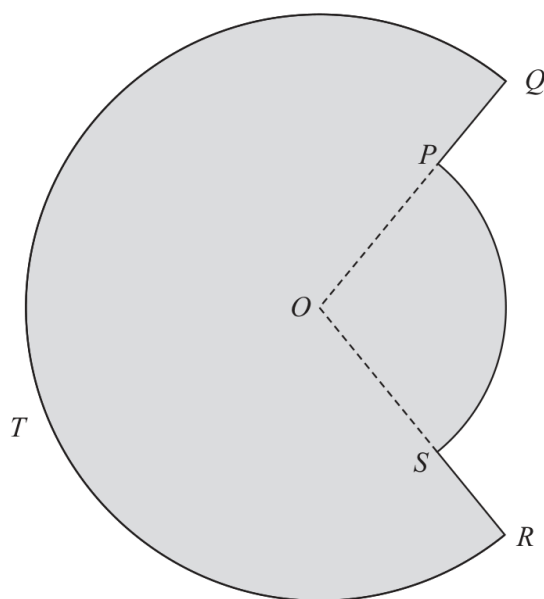


Figure 2

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O .

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8 \text{ m}$

- (a) show that, to 3 significant figures, $OQ = 8.83 \text{ m}$ (3)
- (b) Find the total surface area of the swimming pool in m^2 to the nearest integer. (3)
- (c) Find the total perimeter of the swimming pool in metres to 2 significant figures. (3)

(Total for Question 7 is 9 marks)

Question 4

WMA11/01 October 2025

T10 Radians

F01 Sector measures and inverse sector problems

V03 Concentric, annular and linked sectors

1

Find OP from the small sectorLet $OP = r$. The area of sector POS is 30, so $\frac{1}{2}r^2(1.65) = 30$.

2

Calculate OP $r^2 = \frac{60}{1.65}$, so $r = 6.030... \text{ m}$.

3

Show OQ $OQ = OP + PQ = 6.030... + 2.8 = 8.830...$, so $OQ = 8.83 \text{ m}$ to 3 significant figures.

4

Major sector angleThe major sector $QORTQ$ has angle $2\pi - 1.65$ radians.

5

Total area $\text{Area} = 30 + \frac{1}{2}(8.830...)^2(2\pi - 1.65) = 210.6...$

6

Round areaThe total surface area is 211 m^2 to the nearest integer.

7

Arc lengthsThe inner arc PS has length $6.030...(1.65) = 9.95...$, and the outer major arc has length $8.830...(2\pi - 1.65) = 40.91...$

8

Add straight sides

The straight sides PQ and SR each have length 2.8 m.

9

Perimeter

$9.95... + 40.91... + 2.8 + 2.8 = 56.46...$, so the perimeter is 56 m to 2 significant figures.

FINAL ANSWER

(a) $OQ = 8.83$ m. (b) 211 m². (c) 56 m.

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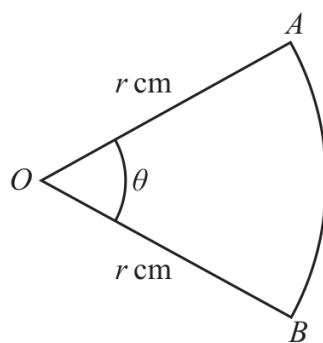


Diagram **NOT**
accurately drawn

Figure 3

Figure 3 shows a sector AOB of a circle, centre O .

The radius OA of the sector is r cm and angle AOB is θ radians.

Given that

- the **area** of sector AOB is 1 cm^2
- the **perimeter** of sector AOB is 8 cm

(a) write down two different equations involving r and θ .

(2)

Given that $r > 2$

(b) solve the two simultaneous equations in part (a) to find the exact value of r and the exact value of θ .

Write each answer in the form $a + b\sqrt{c}$ where a , b and c are integers.

(Solutions relying on calculator technology are not acceptable.)

(5)

(Total for Question 7 is 7 marks)

Question 5

WMA11/01 May/June 2026

T10 Radians

F01 Sector measures and inverse sector problems

V04 Coupled Sector Area-and-Perimeter Constraints

1

(a) Translate the area and perimeterFor a sector of radius r and angle θ radians,

$$\frac{1}{2}r^2\theta = 1.$$

Its perimeter is two radii plus the arc length, so

$$2r + r\theta = 8.$$

2

(b) Eliminate the angle

From the area equation,

$$\theta = \frac{2}{r^2}.$$

Substitute this into the perimeter equation:

$$2r + r\left(\frac{2}{r^2}\right) = 8.$$

Therefore

$$2r + \frac{2}{r} = 8,$$

and, since $r > 0$,

$$r^2 - 4r + 1 = 0.$$

3

(b) Solve for the radius and apply the condition

Complete the square:

$$(r - 2)^2 = 3,$$

so

$$r = 2 \pm \sqrt{3}.$$

The condition $r > 2$ selects

$$r = 2 + \sqrt{3}.$$

4

(b) Find the exact angleUsing $\theta = \frac{2}{r^2}$,

$$\theta = \frac{2}{(2 + \sqrt{3})^2}.$$

Since $\frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3}$,

$$\theta = 2(2 - \sqrt{3})^2 = 2(7 - 4\sqrt{3}) = 14 - 8\sqrt{3}.$$

FINAL ANSWER

$$(a) \frac{1}{2}r^2\theta = 1, \quad 2r + r\theta = 8. \quad (b) \quad r = 2 + \sqrt{3} \text{ cm}, \quad \theta = 14 - 8\sqrt{3} \text{ radians.}$$

Recovering the central angle and choosing the correct arc

WHAT THIS FAMILY ASKS

Obtain the radian angle from a diagram before applying sector formulae, and decide whether the required region uses θ or 2π minus θ .

VARIANT ROUTE

V02 Angle from triangle trigonometry

2 Complete Classified questions

19 total marks

5.

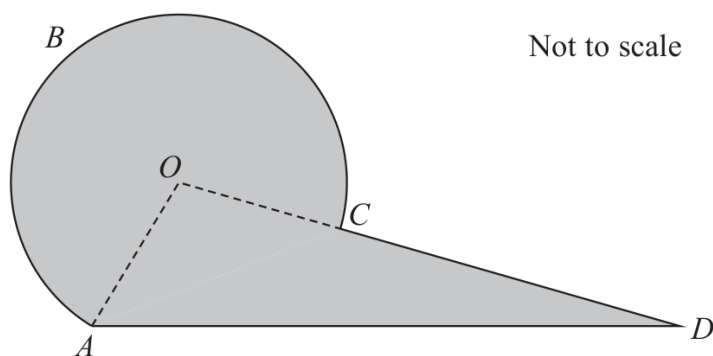


Figure 3

Figure 3 shows the plan view of a viewing platform at a tourist site.

The shape of the viewing platform consists of a sector $ABCOA$ of a circle, centre O , joined to a triangle AOD .

Given that

- $OA = OC = 6$ m
- $AD = 14$ m
- angle $ADC = 0.43$ radians
- angle AOD is an obtuse angle
- OCD is a straight line

find

- (a) the size of angle AOD , in radians, to 3 decimal places, (3)
- (b) the length of arc ABC , in metres, to one decimal place, (2)
- (c) the total area of the viewing platform, in m^2 , to one decimal place. (4)

(Total 9 marks)

Question 6

WMA11/01 January 2021

T10 Radians

F02 Recovering the central angle and choosing the correct arc

V02 Angle from triangle trigonometry

1

Use the sine rule in triangle AOD

Since O, C, D are in a straight line, $\angle ADO = 0.43$. By the sine rule,

$$\frac{14}{\sin \angle AOD} = \frac{6}{\sin 0.43}, \text{ so } \sin \angle AOD = \frac{14 \sin 0.43}{6}.$$

2

Choose the obtuse angle

The calculator gives the acute value $1.337 \dots$, but $\angle AOD$ is obtuse. Hence
 $\angle AOD = \pi - 1.337 \dots = 1.805 \dots$, so $\angle AOD = 1.805$ radians.

3

Find the major-sector angle

Arc ABC is the major arc, so its angle is $2\pi - 1.805 \dots = 4.478 \dots$ radians.

4

Find the arc length

$s = r\theta = 6(4.478 \dots) = 26.868 \dots$, so the length of arc ABC is **26.9 m**.

5

Find the sector area

The sector area is $\frac{1}{2}r^2\theta = \frac{1}{2}(6)^2(4.478 \dots) = 80.607 \dots$

6

Find the triangle area and total area

$\angle OAD = \pi - 0.43 - 1.805 \dots = 0.9066 \dots$ The triangle area is

$$\frac{1}{2}(6)(14) \sin(0.9066 \dots) = 33.093 \dots$$
 Total area = $80.607 \dots + 33.093 \dots = 113.7 \text{ m}^2$.

FINAL ANSWER

(a) 1.805 rad. (b) 26.9 m. (c) 113.7 m².

7.

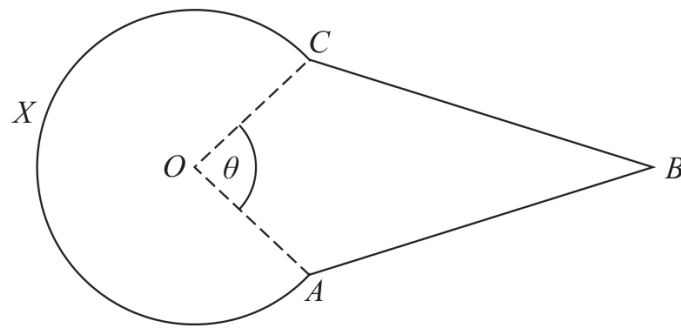


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O .

In the design

- $OA = OC = 0.6$ m
- $AB = CB = 1.4$ m
- Angle $OAB = \text{Angle } OCB = 2$ radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

- (a) show that $\theta = 1.64$ radians to 3 significant figures, (4)
- (b) find the perimeter of the sign, in metres to 2 significant figures, (2)
- (c) find the area of the sign, in m^2 to 2 significant figures. (4)

(Total 10 marks)

Question 7

WMA11/01 October 2021

T10 Radians

F02 Recovering the central angle and choosing the correct arc

V02 Angle from triangle trigonometry

1

Work in triangle OABIn triangle OAB , $OA = 0.6$, $AB = 1.4$, and $\angle OAB = 2$ radians.

2

Find OBBy the cosine rule, $OB^2 = 0.6^2 + 1.4^2 - 2(0.6)(1.4)\cos 2$, so $OB = 1.73756 \dots$

3

Find half of thetaUsing the cosine rule again, $\cos \angle AOB = \frac{0.6^2 + OB^2 - 1.4^2}{2(0.6)(OB)}$, giving $\angle AOB = 0.822199 \dots$ radians.

4

Use symmetry of the kiteThe kite is symmetric, so $\theta = 2\angle AOB = 1.644398 \dots$, hence $\theta = 1.64$ radians to 3 significant figures.

5

Find the outside arc lengthThe sector $OCXA$ uses the major angle $2\pi - \theta$. Its arc length is $0.6(2\pi - 1.644398 \dots) = 2.78327 \dots$

6

Find the perimeterPerimeter = $AB + CB + \text{major arc} = 1.4 + 1.4 + 2.78327 \dots = 5.58327 \dots$, so the perimeter is 5.6 m.

7

Find the kite area

The kite consists of two congruent triangles like OAB , so area of kite = $2 \left(\frac{1}{2} (0.6)(1.4) \sin 2 \right) = 0.76381 \dots$

8

Find the sector area and total area

Major sector area = $\frac{1}{2} (0.6)^2 (2\pi - 1.644398 \dots) = 0.83498 \dots$ Total area = $0.76381 \dots + 0.83498 \dots = 1.59879 \dots$, so the area is 1.6 m^2 .

FINAL ANSWER

(a) $\theta = 1.64 \text{ rad}$. (b) 5.6 m . (c) 1.6 m^2 .

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Composite sector designs

WHAT THIS FAMILY ASKS

Assemble or subtract sector, triangle, trapezium, rectangle and radial-edge contributions to find a complete area or perimeter.

VARIANT ROUTE

V01 Minor sector joined to one or more polygons

V02 Major sector with an attached or removed polygon

V03 Repeated rotational sector design

11 Complete Classified questions

110 total marks

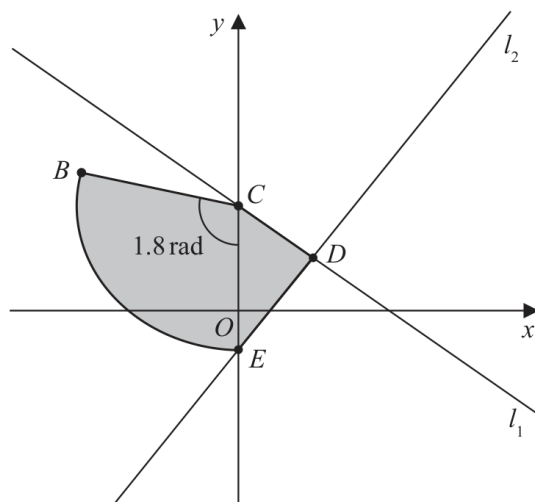


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

- (a) State the y coordinate of C . (1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y -axis at the point E as shown in Figure 3.

- (b) Show that the y coordinate of E is $-\frac{14}{3}$ (3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

- (c) find the length of arc BE . (3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE .

- (d) Calculate the exact area of the region $CBED$. (3)

(Total 10 marks)

Question 8

WMA11/01 May/June 2021

T10 Radians

F03 Composite sector de-
signsV01 Minor sector joined to
one or more polygons

1

Find C

On the y-axis, $x = 0$. From $4y + 3x = 48$, $4y = 48$, so the y-coordinate of C is 12.

2

Find the gradient of l_1

$4y + 3x = 48$ gives $y = -\frac{3}{4}x + 12$, so the gradient of l_1 is $-\frac{3}{4}$.

3

Use perpendicular gradients for l_2

Since l_2 is perpendicular to l_1 , its gradient is $\frac{4}{3}$. Through $D(8, 6)$, $y - 6 = \frac{4}{3}(x - 8)$.

4

Show the y-coordinate of E

$y = \frac{4}{3}x - \frac{32}{3} + 6 = \frac{4}{3}x - \frac{14}{3}$. Therefore when $x = 0$, $E_y = -\frac{14}{3}$.

5

Find the radius of the sector

$C = (0, 12)$ and $E = \left(0, -\frac{14}{3}\right)$, so $CE = 12 + \frac{14}{3} = \frac{50}{3}$.

6

Find arc BE

For sector BCE , $r = \frac{50}{3}$ and $\theta = 1.8 = \frac{9}{5}$. Hence arc $BE = r\theta = \frac{50}{3} \cdot \frac{9}{5} = 30$.

7

Find the sector area

$$\text{Sector area} = \frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{50}{3}\right)^2\left(\frac{9}{5}\right) = 250.$$

8

Find the triangle area and total area

Triangle CDE has vertical base $CE = \frac{50}{3}$ and horizontal height 8, so its area is $\frac{1}{2} \cdot \frac{50}{3} \cdot 8 = \frac{200}{3}$. Total area = $250 + \frac{200}{3} = \frac{950}{3}$.

FINAL ANSWER

$$(a) 12. \quad (b) E_y = -\frac{14}{3}. \quad (c) 30. \quad (d) \frac{950}{3}.$$

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6.

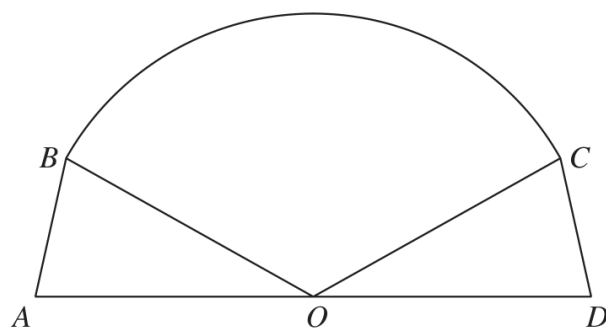


Figure 1

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector BOC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m , to 2 decimal places,

(2)

(b) find the size of angle AOB , in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2 , to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m , to one decimal place.

(4)

(Total for Question 6 is 10 marks)

Question 9

WMA11/01 January 2023

T10 Radians

F03 Composite sector de-
signsV01 Minor sector joined to
one or more polygons

1

Find the radius

Area of sector BOC is $\frac{1}{2}r^2\theta$. So $40 = \frac{1}{2}r^2(2.4) = 1.2r^2$, giving $r = \sqrt{\frac{40}{1.2}} = 5.77$ m.

2

Find angle AOB

The triangles OAB and ODC are congruent, so $\angle AOB = \angle COD$. Since AOD is a straight line, $2\angle AOB + 2.4 = \pi$.

3

Calculate the angle

$\angle AOB = \frac{\pi - 2.4}{2} = 0.3708 \dots$, so 0.37 rad.

4

Find the total area

The two congruent triangle areas together are
 $2\left(\frac{1}{2}(6.25)(5.7735 \dots) \sin 0.3708 \dots\right) = 13.075 \dots$

5

Add the sector area

Total area = $40 + 13.075 \dots = 53.1 \text{ m}^2$ to one decimal place.

6

Find AB

Using the cosine rule, $AB^2 = 6.25^2 + r^2 - 2(6.25)(r) \cos 0.3708 \dots$, giving $AB = 2.265 \dots$ m.

7

Find the perimeter

Perimeter = $AD + AB + CD + \text{arc } BC = 12.5 + 2(2.265 \dots) + 5.7735 \dots (2.4) = 30.9$ m.

FINAL ANSWER

(a) $r = 5.77$ m. (b) 0.37 rad. (c) 53.1 m². (d) 30.9 m.

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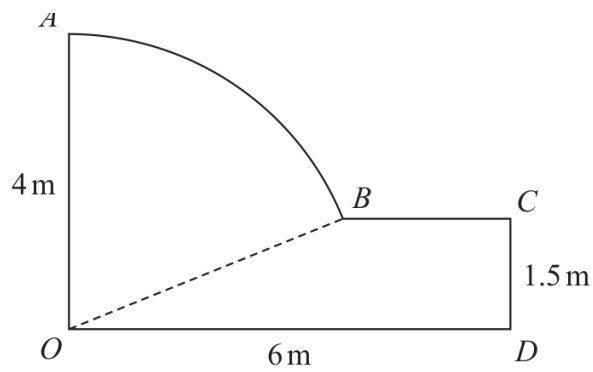


Figure 1

Figure 1 shows the plan for a garden.

In the plan

- OA and CD are perpendicular to OD
- AB is an arc of the circle with centre O and radius 4 metres
- BC is parallel to OD
- OD is 6 metres, OA is 4 metres and CD is 1.5 metres

(a) Show that angle AOB is 1.186 radians to 4 significant figures.

(2)

(b) Find the perimeter of the garden, giving your answer in metres to 3 significant figures.

(4)

(c) Find the area of the garden, giving your answer in square metres to 3 significant figures.

(4)

(Total for Question 5 is 10 marks)

Question 10

WMA11/01 May/June 2023

T10 Radians

F03 Composite sector designs

V01 Minor sector joined to one or more polygons

1

Find the angle AOB

Since $OB = 4$ and the height of B is 1.5, $\cos \angle AOB = \frac{1.5}{4} = 0.375$.

2

Calculate the angle

$\angle AOB = \cos^{-1}(0.375) = 1.186 \dots$ rad, which is 1.186 rad to 4 significant figures.

3

Find BC

The horizontal coordinate of B is $4 \sin(1.186 \dots) = 3.708 \dots$, so $BC = 6 - 3.708 \dots = 2.2919 \dots$ m.

4

Find the arc length

Arc $AB = r\theta = 4(1.186 \dots) = 4.7456 \dots$ m.

5

Find the perimeter

Perimeter = $OA + \text{arc } AB + BC + CD + OD = 4 + 4.7456 \dots + 2.2919 \dots + 1.5 + 6 = 18.537 \dots$ m.

6

Find the area

Area of sector $AOB = \frac{1}{2}(4^2)(1.186 \dots) = 9.491 \dots$

7

Add the quadrilateral area

Area of $OBCD$ is the trapezium area $\frac{1}{2}(6 + 2.2919 \dots)(1.5) = 6.2189 \dots$ **Total area** $= 15.710 \dots = 15.7 \text{ m}^2$.

FINAL ANSWER

(a) $\angle AOB = 1.186 \text{ rad}$. (b) 18.5 m . (c) 15.7 m^2 .

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Diagram **NOT**
accurately drawn

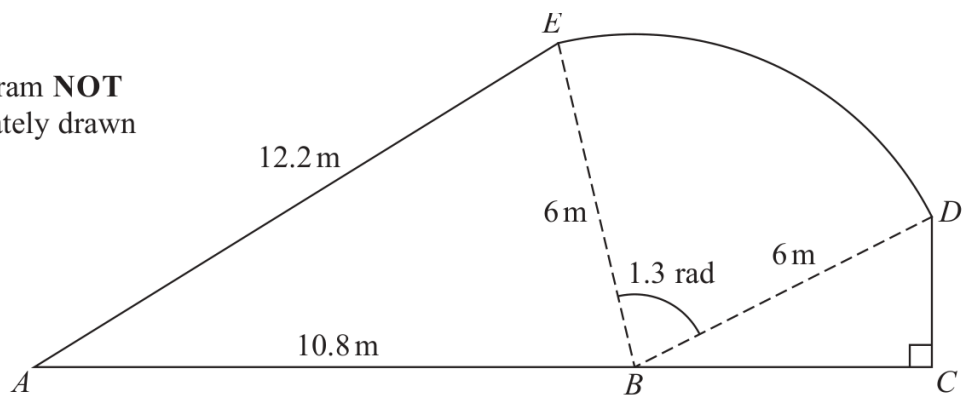


Figure 2

Figure 2 shows the plan view of a garden.

The shape of the garden $ABCDEA$ consists of a triangle ABE and a right-angled triangle BCD joined to a sector BDE of a circle with radius 6 m and centre B .

The points A , B and C lie on a straight line with $AB = 10.8$ m

Angle $BCD = \frac{\pi}{2}$ radians, angle $EBD = 1.3$ radians and $AE = 12.2$ m

- (a) Find the area of the sector BDE , giving your answer in m^2 (2)
- (b) Find the size of angle ABE , giving your answer in radians to 2 decimal places. (2)
- (c) Find the area of the garden, giving your answer in m^2 to 3 significant figures. (3)

(Total for Question 5 is 7 marks)

Question 11

WMA11/01 May/June 2024

T10 Radians

F03 Composite sector designs

V01 Minor sector joined to one or more polygons

1

Find the sector area

$$\text{Area of sector } BDE = \frac{1}{2}(6^2)(1.3) = 23.4 \text{ m}^2.$$

2

Use the cosine rule in triangle ABE

$$12.2^2 = 10.8^2 + 6^2 - 2(10.8)(6) \cos \angle ABE.$$

3

Find angle ABE

$$\cos \angle ABE = \frac{10.8^2 + 6^2 - 12.2^2}{2(10.8)(6)}, \text{ so } \angle ABE \approx 1.54 \text{ radians.}$$

4

Find the area of triangle ABE

$$\text{Area } ABE = \frac{1}{2}(10.8)(6) \sin(1.54147...) \approx 32.386.$$

5

Find angle DBC

$$\angle DBC = \pi - 1.3 - \angle ABE \approx 0.30012 \text{ radians.}$$

6

Find the area of triangle BCD

$$\text{Area } BCD = \frac{1}{2}(6 \cos \angle DBC)(6 \sin \angle DBC) \approx 5.084.$$

7

Add the three areas $23.4 + 32.386 + 5.084 \approx 60.9 \text{ m}^2$ to 3 significant figures.**FINAL ANSWER**

(a) 23.4 m^2 . (b) 1.54 radians . (c) 60.9 m^2 .

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5. A plot of land OAB is in the shape of a sector of a circle with centre O .

Given

- $OA = OB = 5 \text{ km}$
- angle $AOB = 1.2 \text{ radians}$

(a) find the perimeter of the plot of land.

(2)

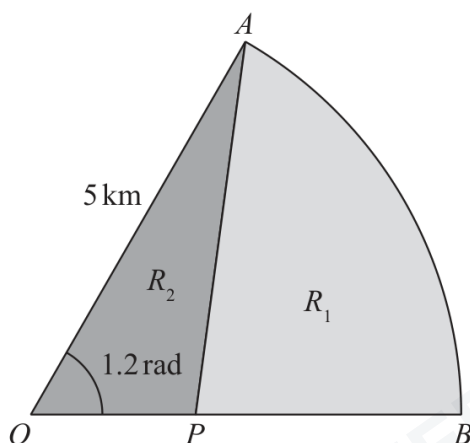


Diagram **NOT**
accurately drawn

Figure 2

A point P lies on OB such that the line AP divides the plot of land into two regions R_1 and R_2 as shown in Figure 2.

Given that

$$\text{area of } R_1 = 3 \times \text{area of } R_2$$

(b) show that the area of $R_2 = 3.75 \text{ km}^2$

(3)

(c) Find the length of AP , giving your answer to the nearest 100 m.

(4)

(Total for Question 5 is 9 marks)

Question 12

WMA11/01 October 2024

T10 Radians

F03 Composite sector designs

V01 Minor sector joined to one or more polygons

1

Find the arc length

$$\text{arc } AB = r\theta = 5(1.2) = 6 \text{ km.}$$

2

Find the perimeter

$$OA + OB + \text{arc } AB = 5 + 5 + 6 = 16 \text{ km.}$$

3

Find the total sector area

$$\text{Area of } OAB = \frac{1}{2}(5^2)(1.2) = 15 \text{ km}^2.$$

4

Use the area ratio

Since area $R_1 = 3 \times$ area R_2 , the total area is $4 \times$ area R_2 .

5

Find area R_2

$$\text{area } R_2 = \frac{15}{4} = 3.75 \text{ km}^2.$$

6

Find OP from triangle OAP

$$\frac{1}{2}(5)(OP) \sin 1.2 = 3.75, \text{ so } OP \approx 1.609 \text{ km.}$$

7

Use the cosine rule

$$AP^2 = 5^2 + OP^2 - 2(5)(OP) \cos 1.2, \text{ so } AP \approx 4.665 \text{ km.}$$

8

Round to the nearest 100 m $AP = 4.7$ km to the nearest 0.1 km.**FINAL ANSWER** (a) 16 km. (b) 3.75 km^2 . (c) $AP = 4.7$ km.

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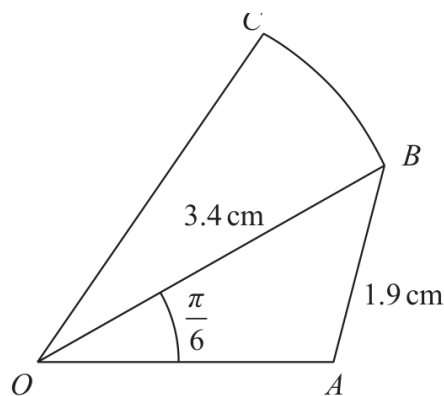


Figure 1

Figure 1 shows a sketch of a design for a badge.

The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

- $OB = 3.4$ cm
- $AB = 1.9$ cm
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(a) find the size of angle OAB , giving your answer in radians to 4 significant figures, (3)

(b) find the area of triangle OAB , in cm^2 , giving your answer to 3 significant figures. (2)

Given that the ratio of the area of sector OBC to the area of triangle OAB is 3 : 2

(c) show that angle BOC is 0.462 radians to 3 significant figures. (3)

(d) Hence find the perimeter of the badge, in cm, to the nearest integer. (5)

(Total for Question 8 is 13 marks)

Question 13

WMA11/01 January 2025

T10 Radians

F03 Composite sector de-signs

V01 Minor sector joined to one or more polygons

1

Use the sine ruleIn triangle OAB , $AB = 1.9$, $OB = 3.4$ and $\angle AOB = \frac{\pi}{6}$.

2

Find sine of angle OAB

$$\frac{\sin \angle OAB}{3.4} = \frac{\sin(\pi/6)}{1.9}, \text{ so } \sin \angle OAB = \frac{17}{19}.$$

3

Choose the obtuse angleSince $\angle OAB > \frac{\pi}{2}$, $\angle OAB = \pi - \sin^{-1}\left(\frac{17}{19}\right) \approx 2.034$ radians.

4

Find OAUsing the sine rule again, $OA \approx 2.096$ cm.

5

Find the triangle area

$$\text{Area } OAB = \frac{1}{2}(OA)(OB) \sin(\pi/6) \approx 1.78 \text{ cm}^2.$$

6

Use the area ratio

$$\text{Area sector } OBC : \text{Area triangle } OAB = 3 : 2, \text{ so sector area} = \frac{3}{2}(1.78156\dots).$$

7

Find angle BOC

$$\frac{1}{2}(3.4)^2\theta = \frac{3}{2}(1.78156\dots), \text{ so } \theta \approx 0.462 \text{ radians.}$$

8

Find the perimeter

The perimeter is $OA + AB + OC + \text{arc } BC = 2.096 + 1.9 + 3.4 + 3.4(0.46234...) \approx 8.968 \text{ cm}$.

9

Round

The perimeter is 9 cm to the nearest integer.

FINAL ANSWER

(a) $\angle OAB = 2.034 \text{ radians}$. (b) 1.78 cm^2 . (c) $\angle BOC = 0.462 \text{ radians}$. (d) 9 cm.

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4.

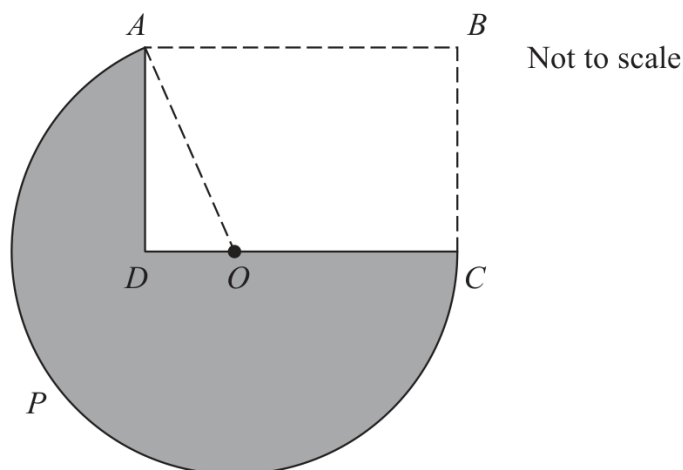


Figure 2

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$ABCD$ is a rectangle with $AB = 16$ m.

$APCOA$ is a sector of a circle centre O with radius 12 m.

The point O lies on the line DC , as shown in Figure 2.

- (a) Show that the size of angle AOD is 1.231 radians to 3 decimal places. (2)

The lawn $APCDA$ is shown shaded in Figure 2.

- (b) Find the area of the lawn, in m^2 , to one decimal place. (4)

- (c) Find the perimeter of the lawn, in metres, to one decimal place. (3)

(Total 9 marks)

Question 14

WMA11/01 January 2020

T10 Radians

F03 Composite sector designs

V02 Major sector with an attached or removed polygon

1

Find angle AOD

Since $OC = 12$ and $DC = AB = 16$, $OD = 4$. In right triangle AOD ,
 $\cos \angle AOD = \frac{OD}{OA} = \frac{4}{12} = \frac{1}{3}$. Hence $\angle AOD = \cos^{-1}\left(\frac{1}{3}\right) = 1.231$ radians to 3 d.p.

2

Identify the major sector

The shaded lawn uses the major sector from A to C , so its angle is $\pi + 1.231$ radians.

3

Subtract the unshaded triangle

$AD = \sqrt{12^2 - 4^2} = 8\sqrt{2}$. Area of major sector = $\frac{1}{2}(12)^2(\pi + 1.231)$. Area of triangle $AOD = \frac{1}{2}(4)(8\sqrt{2})$.

4

Find the lawn area

Area = $\frac{1}{2}(12)^2(\pi + 1.231) - \frac{1}{2}(4)(8\sqrt{2}) = 292.2 \text{ m}^2$ to 1 d.p.

5

Find the lawn perimeter

Perimeter = major arc + CD + $AD = 12(\pi + 1.231) + 16 + 8\sqrt{2} = 79.8 \text{ m}$ to 1 d.p.

FINAL ANSWER

(a) $\angle AOD = 1.231 \text{ rad}$. (b) 292.2 m^2 . (c) 79.8 m .

8.

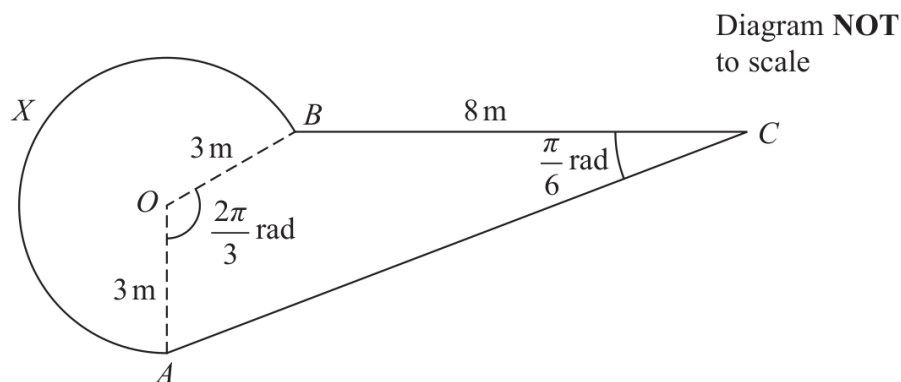


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8 \text{ m}$
- $OA = OB = 3 \text{ m}$
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

(b) Calculate the exact area of the triangle AOB .

(2)

(c) Show that the length AB is $3\sqrt{3} \text{ m}$.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

(Total for Question 8 is 14 marks)

Question 15

WMA11/01 October 2022

T10 Radians

F03 Composite sector designs

V02 Major sector with an attached or removed polygon

1

Find the angle of the sector AOBX

The sector AOBX is the major sector, so its angle is $2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}$.

2

Find the sector area

$$\text{Area} = \frac{1}{2}r^2\theta = \frac{1}{2}(3^2)\left(\frac{4\pi}{3}\right) = 6\pi \text{ m}^2.$$

3

Find the sector perimeter

The arc length is $r\theta = 3\left(\frac{4\pi}{3}\right) = 4\pi$. Adding the two radii gives $4\pi + 6 \text{ m}$.

4

Find the area of triangle AOB

$$\text{Area} = \frac{1}{2}(3)(3)\sin\left(\frac{2\pi}{3}\right) = \frac{9}{2} \cdot \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{4} \text{ m}^2.$$

5

Show AB

By the cosine rule, $AB^2 = 3^2 + 3^2 - 2(3)(3)\cos\left(\frac{2\pi}{3}\right) = 18 + 9 = 27$, so $AB = 3\sqrt{3}$.

6

Find AC

In triangle ABC, $AB^2 = AC^2 + 8^2 - 2(AC)(8)\cos\left(\frac{\pi}{6}\right)$, so $27 = AC^2 - 8\sqrt{3}AC + 64$.

7

Choose the correct length

Solving gives $AC = 4\sqrt{3} \pm \sqrt{11}$. From the diagram, AC is the longer value, so $AC = 4\sqrt{3} + \sqrt{11}$.

8

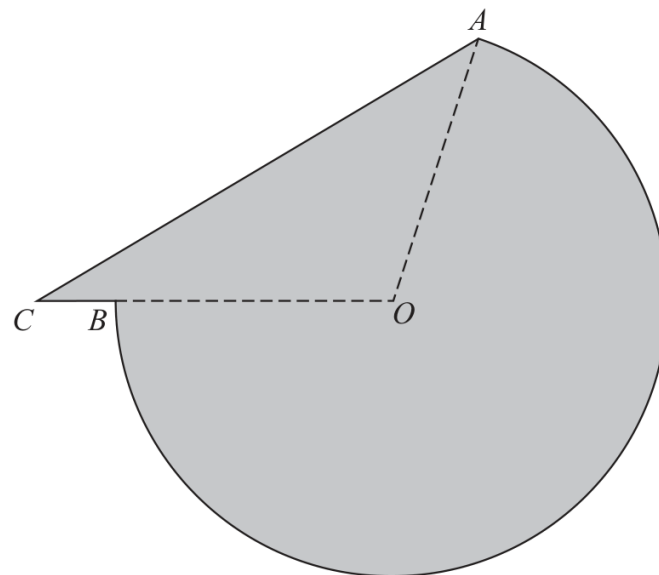
Find the total area

Area of triangle ABC is $\frac{1}{2}(8)(AC) \sin\left(\frac{\pi}{6}\right) = 2AC = 8\sqrt{3} + 2\sqrt{11}$. Total area is $6\pi + \frac{9\sqrt{3}}{4} + 8\sqrt{3} + 2\sqrt{11} = 43.236 \dots$, so 43 m^2 to 2 significant figures.

FINAL ANSWER

(a)(i) $6\pi \text{ m}^2$, (ii) $4\pi + 6 \text{ m}$. (b) $\frac{9\sqrt{3}}{4} \text{ m}^2$. (c) $AB = 3\sqrt{3} \text{ m}$.
(d) 43 m^2 .

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Not to scale

Figure 2

The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector AOB of a circle centre O joined to a triangle AOC .

Given that

- $AO = OB = 15$ m
- $BC = 2$ m
- CBO is a straight line
- angle $ACO = 0.6$ radians

(a) show that angle COA is 1.847 radians to 3 decimal places.

(3)

(b) Find the total area of the helicopter landing pad.
Give your answer in m^2 to 3 significant figures.

(3)

(c) Find the perimeter of the helicopter landing pad.
Give your answer in metres to 3 significant figures.

(3)

(Total for Question 6 is 9 marks)

Question 16

WMA11/01 May/June 2025

T10 Radians

F03 Composite sector designs

V02 Major sector with an attached or removed polygon

1

Set up triangle AOC $AO = 15$, $OC = OB + BC = 17$, and $\angle ACO = 0.6$.

2

Use the sine rule $\frac{\sin \angle CAO}{17} = \frac{\sin 0.6}{15}$, so $\angle CAO \approx 0.6944$ radians.

3

Show angle COA $\angle COA = \pi - 0.6 - 0.6944... = 1.847...$, so $\angle COA = 1.847$ radians to 3 decimal places.

4

Major sector angleThe shaded sector is the major sector AOB , so its angle is $2\pi - 1.847...$

5

Sector areaSector area $= \frac{1}{2}(15)^2(2\pi - 1.847...) = 499.15...$

6

Triangle areaArea $AOC = \frac{1}{2}(15)(17) \sin(1.847...) = 122.73...$

7

Total area $499.15... + 122.73... = 621.88...$, so the total area is 622 m^2 to 3 significant figures.

8

Find AC

By the cosine rule, $AC^2 = 15^2 + 17^2 - 2(15)(17) \cos(1.847\dots)$, so $AC \approx 27.57$ m.

9

Major arc length

arc $AB = 15(2\pi - 1.847\dots) = 66.54\dots$ m.

10

Perimeter

$66.54\dots + 27.57\dots = 94.11\dots$, so the perimeter is 94.1 m.

FINAL ANSWER

(a) $\angle COA = 1.847$ rad. (b) 622 m^2 . (c) 94.1 m.

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5.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

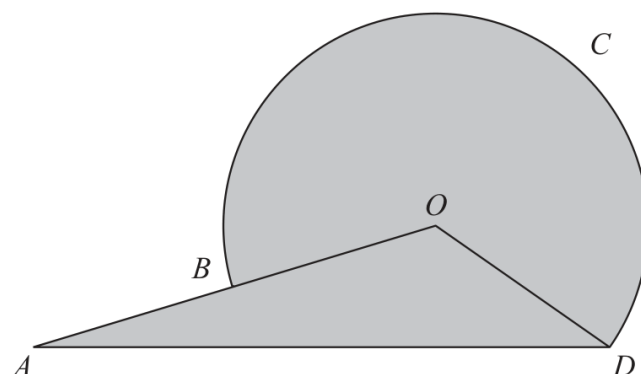


Diagram NOT
to scale

Figure 2

Figure 2 shows the logo for the tourist board of an island.

The logo consists of a sector $BODCB$ of a circle centre O joined to a triangle AOD .

Given that

- ABO is a straight line
- $OD = 3$ cm
- $AD = 8$ cm
- angle $AOD = 2.5$ radians

(a) show that angle ODA is 0.415 radians to 3 significant figures.

(3)

(b) Find the total area of the logo, in cm^2 , to one decimal place.

(3)

(c) Find the perimeter of the logo, $ABCD A$, in cm, to one decimal place.

(3)

(Total for Question 5 is 9 marks)

Question 17

WMA11/01 January 2026

T10 Radians

F03 Composite sector designs

V02 Major sector with an attached or removed polygon

1

Use triangle AODIn triangle AOD , $OD = 3$, $AD = 8$ and $\angle AOD = 2.5$ radians.

2

Use the sine rule

$$\frac{\sin \angle OAD}{3} = \frac{\sin 2.5}{8}, \text{ so } \angle OAD = \sin^{-1} \left(\frac{3 \sin 2.5}{8} \right) = 0.2263\dots$$

3

Show angle ODA
 $\angle ODA = \pi - 2.5 - 0.2263\dots = 0.4152\dots$, so $\angle ODA = 0.415$ radians to 3 significant figures.

4

Area of the major sector
 The sector has radius 3 and angle $2\pi - 2.5$, so its area is $\frac{1}{2}(3)^2(2\pi - 2.5) = 17.02\dots$

5

Area of triangle AOD

$$\text{Area of triangle } AOD = \frac{1}{2}(3)(8) \sin 0.4152\dots = 4.84\dots$$

6

Total area
 $17.02\dots + 4.84\dots = 21.86\dots$, so the area is 21.9 cm^2 to one decimal place.

7

Find AO
 By the cosine rule, $AO^2 = 3^2 + 8^2 - 2(3)(8) \cos 0.4152\dots$, so $AO \approx 5.39 \text{ cm}$.

8

Find AB

Since A, B, O are collinear and $OB = 3$, $AB = AO - OB \approx 5.39 - 3 = 2.39$ **cm**.

9

Perimeter

$AB + \text{major arc } BD + AD = 2.39... + 3(2\pi - 2.5) + 8 = 21.7$ **cm to one decimal place**.

FINAL ANSWER

(a) $\angle ODA = 0.415$ **rad**. (b) 21.9 **cm²**. (c) 21.7 **cm**.

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8.

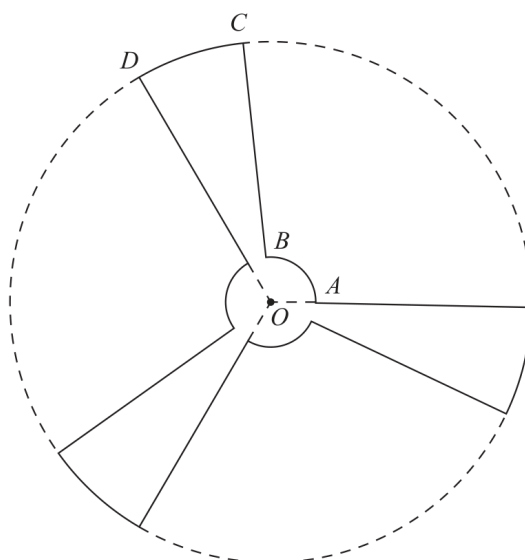


Figure 3

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to $OABCO$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

- (a) show that the length of the arc CD is 35.9 cm to one decimal place.

(3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

- (b) the perimeter of the face of the fan, giving your answer to the nearest cm,

(2)

- (c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear.

(5)

(Total 10 marks)

Question 18

WMA11/01 May/June 2022

T10 Radians

F03 Composite sector design

V03 Repeated rotational sector design

1

Find the small sector angleFor arc AB , $s = r\theta$, so $15 = 9\theta$ and $\theta = \frac{5}{3}$ radians.

2

Find the outer arc angleEach identical section has total angle $\frac{2\pi}{3}$, so the angle for arc CD is $\frac{2\pi}{3} - \frac{5}{3} = \frac{2\pi - 5}{3}$.

3

Find CD $CD = 84 \left(\frac{2\pi - 5}{3} \right) = 28(2\pi - 5) = 35.9$ cm to 3 significant figures.

4

Find the perimeterThere are 3 inner arcs, 3 outer arcs, and 6 straight radial edges of length $84 - 9 = 75$. Thus $P = 3(15) + 3(35.929 \dots) + 6(75) = 602.8 \dots$ cm.

5

Find the area of one sectionOne section has area $\frac{1}{2}(9^2) \left(\frac{5}{3} \right) + \frac{1}{2}(84^2) \left(\frac{2\pi - 5}{3} \right) = 1576.525 \dots$ cm².

6

Find the total areaThe total area is $3(1576.525 \dots) = 4729.57 \dots$ cm² = 4.73×10^3 cm² to 3 significant figures.

FINAL ANSWER

(a) $CD = 35.9$ cm. (b) 603 cm. (c) 4.73×10^3 cm².

Circular segments and chords

WHAT THIS FAMILY ASKS

Use sector-minus-triangle area and arc-plus-chord perimeter for a circular segment.

VARIANT ROUTE

V01 Segment area

1 Complete Classified questions

7 total marks

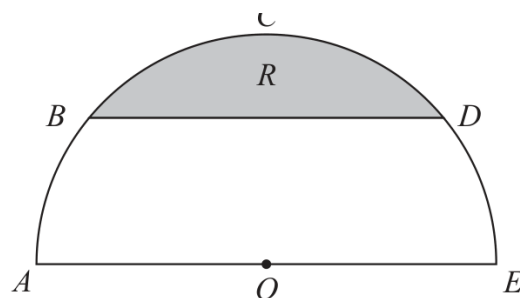


Figure 2

Figure 2 shows a plan view of a semicircular garden $ABCDEOA$

The semicircle has

- centre O
- diameter AOE
- radius 3 m

The straight line BD is parallel to AE and angle BOA is 0.7 radians.

(a) Show that, to 4 significant figures, angle BOD is 1.742 radians.

(1)

The flowerbed R , shown shaded in Figure 2, is bounded by BD and the arc BCD .

(b) Find the area of the flowerbed, giving your answer in square metres to one decimal place.

(3)

(c) Find the perimeter of the flowerbed, giving your answer in metres to one decimal place.

(3)

(Total 7 marks)

Question 19

WMA11/01 January 2022

T10 Radians

F04 Circular segments and V01 Segment area
chords

1

Find angle BOD

Since BD is parallel to the diameter AE , the two side angles at the centre are both 0.7 radians. Therefore $\angle BOD = \pi - 0.7 - 0.7 = \pi - 1.4 = 1.74159 \dots$, so 1.742 radians to 4 significant figures.

2

Find the sector area

With $r = 3$ and $\theta = 1.74159 \dots$, sector area $= \frac{1}{2}r^2\theta = \frac{1}{2}(3)^2(1.74159 \dots) = 7.837 \dots$

3

Subtract triangle BOD

Triangle area $= \frac{1}{2}(3)(3) \sin(1.74159 \dots) = 4.434 \dots$ The segment area is $7.837 \dots - 4.434 \dots = 3.403 \dots$, so 3.4 m^2 .

4

Find the arc length

Arc $BCD = r\theta = 3(1.74159 \dots) = 5.225 \dots$

5

Find the chord and perimeter

$BD = 2r \sin \frac{\theta}{2} = 6 \sin(0.87079 \dots) = 4.590 \dots$ Perimeter $= 5.225 \dots + 4.590 \dots = 9.8 \text{ m}$.

FINAL ANSWER

(a) $\angle BOD = 1.742 \text{ rad}$. (b) 3.4 m^2 . (c) 9.8 m .